Solution

i) 18.4 ii) 19.4 iii) 101.6×10^6
iv) 248000 v) 12.7

1.7.3 Arithmetical Operations with Significant Figures

(i) Addition and subtraction

In addition and subtraction, the final result should retain as many decimal places as there are in the number with the smallest number of decimal places.

Example:

1. $3.1 + 1.780 + 2.046 = 6.926$

   Here the least number of significant digits after the decimal is one. Hence the result will be 6.9.

2. $12.637 - 2.42 = 10.217$

   Here the least number of significant digits after the decimal is two. Hence the result will be 10.22

(ii) Multiplication and Division

In multiplication or division, the final result should retain as many significant figures as there are in the original number with smallest number of significant figures.

Example:

1. $1.21 \times 36.72 = 44.4312 = 44.4$

   Here the least number of significant digits in the measured values is three. Hence the result when rounded off to three significant digits is 44.4

2. $36.72 \div 1.2 = 30.6 = 31$

   Here the least number of significant digits in the measured values is two. Hence the result when rounded off to significant digit becomes 31.

1.8 DIMENSIONAL ANALYSIS

1.8.1 Dimension of Physical Quantities

In mechanics, we deal with the physical quantities like mass, time, length, velocity, acceleration, etc. which can be expressed in terms of three independent base quantities such as M, L and T. So, the dimension of a physical quantity can be defined as 'any physical quantity which is expressed in terms of base quantities whose exponent (power) represents the dimension of the physical quantity'. The notation used to denote the dimension of a physical quantity is [(physical quantity within square bracket)]. For an example, [length] means dimension of length, [area] means dimension of area, etc. The dimension of length can be expressed in terms of base quantities as

$$[\text{length}] = M^0 LT^0 = L$$

Similarly, $[\text{area}] = M^0 L^2 T^0 = L^2$

Similarly, $[\text{volume}] = M^0 L^3 T^0 = L^3$

Note that in all the cases, the base quantity L is same but exponent (power) are different, which means dimensions are different. For a pure number, exponent of base quantity is zero. For example, consider the number 2, which has no dimension and can be expressed as

$$\Rightarrow [2] = M^0 L^0 T^0 \qquad \text{(dimensionless)}$$

Let us write down the dimensions of a few more physical quantities.

$$\text{Speed, } s = \frac{\text{distance}}{\text{time taken}} \Rightarrow [s] = \frac{L}{T} = LT^{-1}$$

$$\text{Velocity, } \vec{v} = \frac{\text{displacement}}{\text{time taken}} \Rightarrow [\vec{v}] = \frac{L}{T} = LT^{-1}$$

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